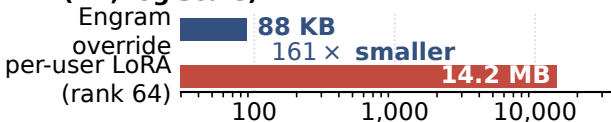


**(a) State for 100 inserted facts
(KB, log scale)**



**(b) Fact lifecycle avoids
adapter retraining**

ENGRAM

WRITE

closed form; OPT

DELETE MAP

drop map

PER-USER LoRA

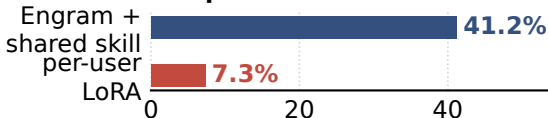
WRITE

train adapter

DELETE FACT

retrain / unlearn

**(c) Local facts support indirect
and multi-hop use**



indirect reasoning over
held-out user facts (%)